KAMAL PATEL

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EDUCATION

Rutgers, The State University of New Jersey · New Jersey *September 2022 - June 2026*
*MS & **Doctoral** studies in Industrial & Systems Engineering (**Completed two years of PhD courses**)*

Courses: Production Analysis, Supply Chain Engineering, Production Analysis, **Lean Six Sigma(Rutgers Business School),** Regression Analysis, Linear Programming, Systems Reliability, Data Mining II, Time-Series Analysis,

Certification: Lean Six Sigma Green Belt (CFQ International LLC)

PROJECTS

Electricity Price Forecasting using Autoregressive and LSTM Model - Forecasting Analytics Course *GitHub*
- Engineered PyTorch LSTM and autoregressive models to forecast Locational Marginal Pricing (LMP), optimizing prediction accuracy through advanced time-series feature engineering.
- Validated model performance against traditional baselines, achieving an MAE of 3.92 and RMSE of 9.21 to outperform the benchmark SARIMAX model.

End-to-End Inventory & Supply Chain Analytics Dashboard | *Power BI (DAX)* *GitHub*
- Developed an interactive Power BI dashboard tracking core supply chain **KPIs,** including Warehouse Utilization Rate, Days Sales of Inventory (DSI), and Inventory Turnover Ratio.
- Built an interactive Power BI dashboard querying multi-warehouse inventory datasets to track SKU turnover, safety stock, and reorder alerts.
- Modeled relational inventory datasets in Power BI to deliver actionable recommendations for inventory replenishment and stocking strategy.

Loan Default Risk Prediction - Data Mining II Course *GitHub*
- Preprocessed a 300K+ record dataset by conducting exploratory data analysis (EDA) in Python to clean missing values, handle anomalies, and treat outliers.
- Engineered XGBoost and LightGBM predictive models to quantify risk factors, achieving a 74.4% AUC score (top percentile in Kaggle benchmark).

WORK EXPERIENCE

Rutgers University - Industrial & Systems Engineering *Jan 2025 – June 2026*
Graduate Teaching Assistant Piscataway, NJ
- Assisted 30+ students on linear programming concepts and mentored optimization projects using Excel Solver.
- Facilitated weekly recitations for reliability engineering course and guided students through MATLAB-based computational assignments and analysis.

Kismet Technologies *May 2023 - Aug 2023*
Industrial Engineer Intern *Orlando, FL*
- Implemented Katana MRP software for inventory management and tracking, driving a **20%** improvement in inventory accuracy.
- Revamped batch manufacturing data tracking by building customized PowerApps application, boosting operational efficiency by **11%.**
- Initiated production schedules and authored standard operating procedures (SOPs) for batch acceptance testing and to ensure ISO 9001 compliance.

Mareana: Industry Consulting *Nov 2022 - Dec 2022*
Project Head *New Brunswick, NJ*
- Spearheaded a 4-person consulting team analyzing production line efficiency for a turbine manufacturer.
- Engineered a **FlexSim discrete event model** to optimize line balancing, reducing production cycle time by **23.5%.**
- Analyzed production data and proposed a solution with a projected throughput increase by **10–12%.**

Susha Founders and Engineers *Dec 2021 - May 2022*
Manufacturing Engineer Intern *Gujarat, India*
- Conducted time-and-motion studies across assembly stages for 2 production processes, reducing cycle time by 8%.
- Analyzed historical inventory datasets using SQL, VLOOKUPs, and Pivot Tables to forecast demand and establish new EOQ and reorder points.
- Enforced 5S principles and optimized monthly Overall Equipment Effectiveness (OEE) tracking, yielding a ~10% operational efficiency gain.

ACHIEVEMENTS/CERTIFICATIONS

- **Data Mining II –** Rutgers University (Best Presentation)
- **Regression Analysis –** Rutgers University (Best Project Performance)
- **SQL for Data Science -** University of California, Davis (Certificate)
- **Predictive Analytics for Decision-Making Specialization** - University of Minnesota

TECHNICAL SKILLS & LEADERSHIP

Data Analytics Tools/Packages	Python(Pandas, NumPy, Scikit-learn), PyTorch, TensorFlow, R, MATLAB , SQL, Minitab
Software Packages	ANSYS, AutoCAD, Flexsim, Gurobi, AnyLogic, SolidWorks, PTC Creo, MS Project, Visio
Operational Excellence	Lean Manufacturing, VSM, Six Sigma (DMAIC), Kaizen, Root Cause Analysis
Leadership	Lead Mechanical Engineer NYU's autonomous vehicle project (**International Ground Vehicle Competition**) - Managed hardware procurement, assembly, and integration